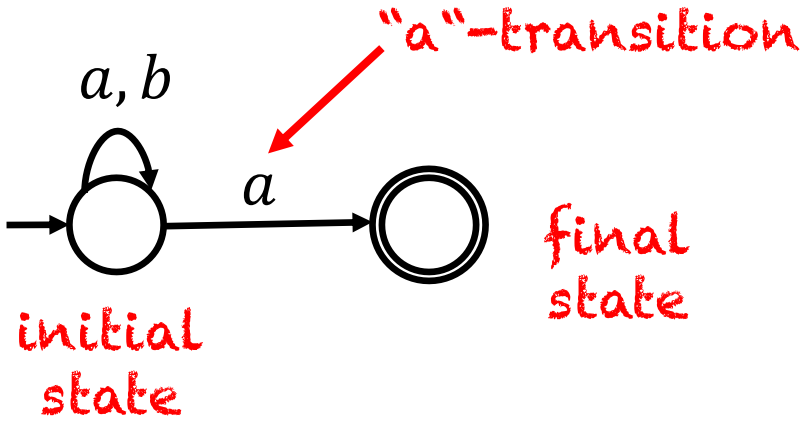
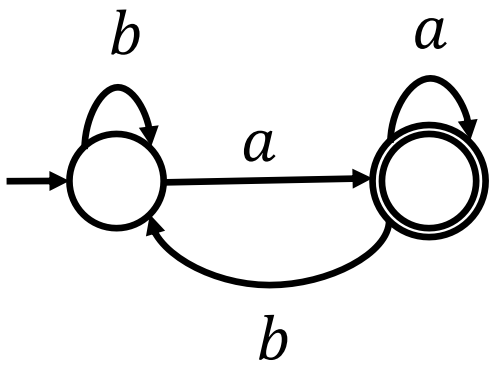
Last week: finite automata

NFA



DFA



- Is the language empty?
- Is the word $ababa$ accepted by the automaton?
- What is the accepted language?

Why are we doing this?

Theoretical computer science ≈ study of mathematical formalisms for languages

- What languages can be described by a formalism? What problems can a computer solve?
- For which problems and classes of languages are there (efficient) algorithms?

This course: *a tour through common formalisms in computer science*

Examples:

- ε, 0, 1, 111, 10101001 are words over the alphabet $\{0,1\}$
- $\{0, 11, 00101, 100\}$ is a language over the alphabet $\{0,1\}$
- *"set of all computer programs that eventually stop on all inputs"* is a language over $\{0,1\}$
- Membership problem: does a word belong to a language?
- For regular languages described by DFAs, we can efficiently solve membership
- For *arbitrary computer programs*, membership cannot be solved at all!

Formal Recap: Words

Alphabet Σ: finite set of *symbols*, e.g. $a, b, c, 0, 1, \ldots$

The set of words over Σ is defined inductively as follows:

- The empty word ε is a word over Σ
- Every symbol a in Σ is a word over Σ
- If u is a word over Σ and $a \in \Sigma$, then $a\,u$ and $u\,a$ are words over Σ

The length of a word w is given by $|w| = \begin{cases} 0, & \text{if } w = \varepsilon \\ 1 + |u|, & \text{if } w = a\,u \text{ and } a \in \Sigma \end{cases}$

The concatenation of v and w is $v \cdot w = v\,w = \begin{cases} v, & \text{if } w = \varepsilon \\ (v\,a) \cdot z, & \text{if } w = a\,u \text{ and } a \in \Sigma \end{cases}$

For $v = a_1 \ldots a_n$ and $w = b_1 \ldots b_m$, we have $v \cdot w = a_1 \ldots a_n\, b_1 \ldots b_m$

Formal Recap: Languages

Language L: set of words over Σ

$L_1\ L_2$: concatenation of L_1 and L_2

$$L_1 L_2 = L_1 \cdot L_2 = \{\, u\, v \mid u \in L_1, v \in L_2 \}$$

L^k: k concatenated copies of L

- $L^0 = \{\, \varepsilon \,\}$
- $L^{k+1} = L\, L^k$

Kleene closure of L: $L^* = \bigcup_{\{k \geq 0\}} L^k$

Examples over $\Sigma = \{a, b\}$:

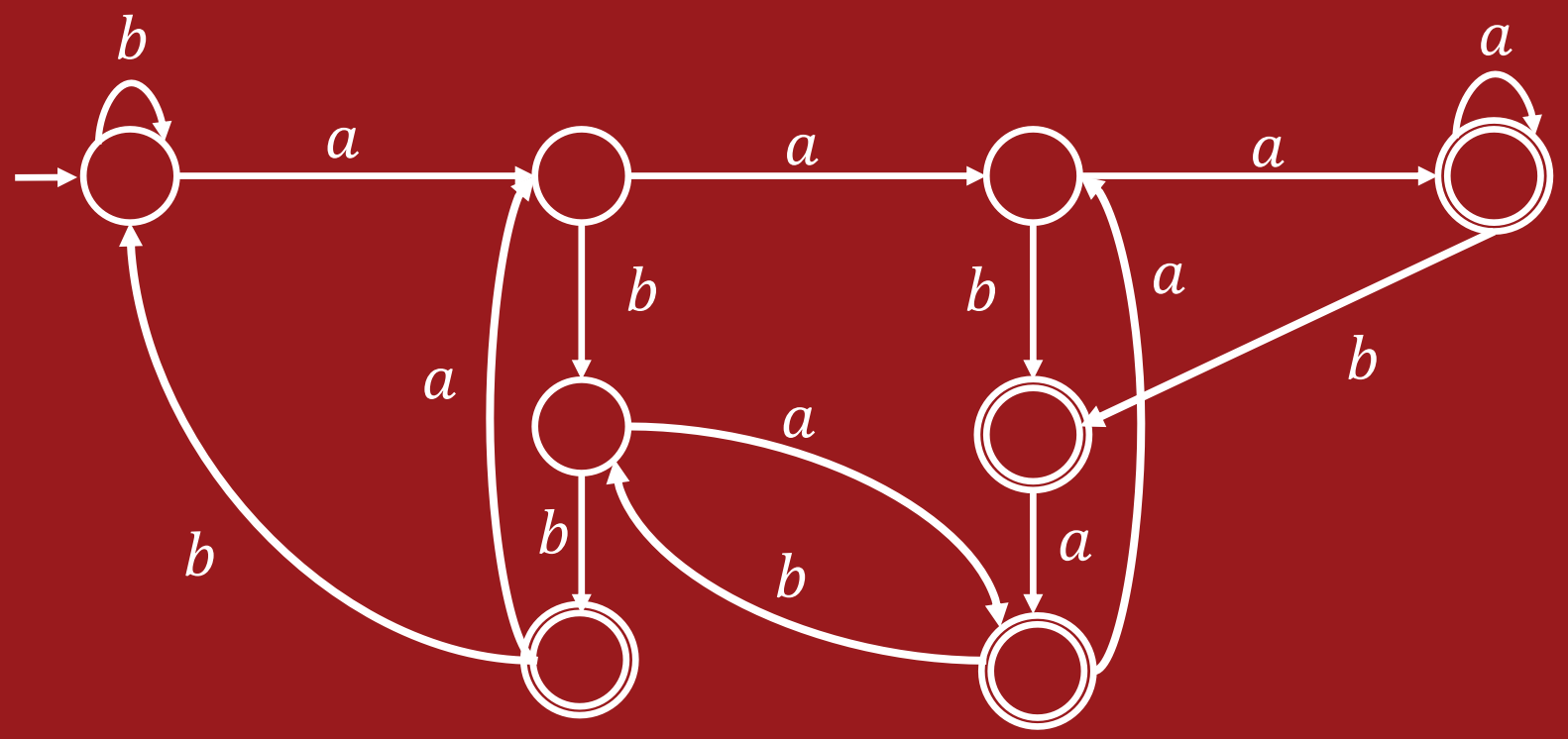
$L_1 = \{ab\}$
$L_2 = \{ababab\}$
$L_3 = \{b, ab, aab, aaab, aaaab, \ldots\}$

$L_2 = L_1^3 = L_1\ L_1\ L_1$

$L_3 = \{a\}^*\ \{\, b \,\}$

What languages are given by $\emptyset^0$ $\emptyset^3$ $\emptyset^*$ $\{\varepsilon\}^0$ $\{\varepsilon\}^5$ $\{\varepsilon\}^*$?

Another example: what is the accepted language?



Two ways to describe a formal language

Automata (DFA, NFA)

- Easy to check membership
- We can construct automata for union, intersection, complement, …
- Verbose: might require many states, even for simple languages

Declarative (mathematics)

- Concise, often more readable
- More expressive than automata
- Unclear how to check membership

➔ Find mathematical notation for the languages accepted by finite automata

02141 – Computer Science Modelling

RL3 – REGULAR EXPRESSIONS

Agenda

1. What are regular expressions?
2. Applications
3. Regular Expressions vs. Finite Automata
4. Properties of Regular Expressions

What are regular expressions?

“concise algebraic notation for regular languages”

Example

language union

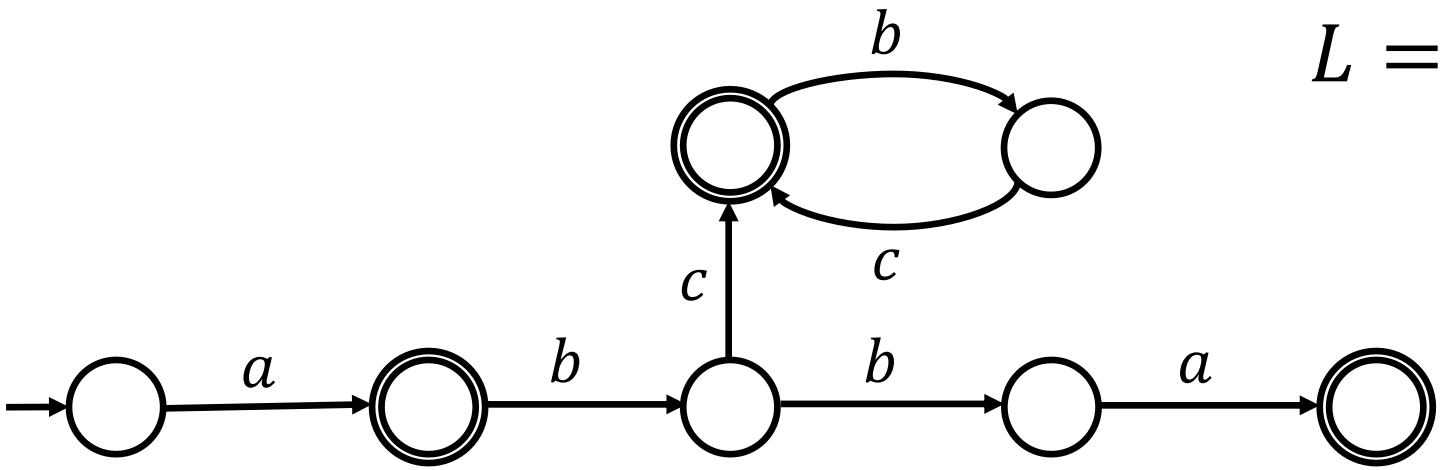
Kleene closure

$$abba + a \cdot (bc)^*$$

every word is a regular expression

language concatenation

$$L = \{ abba \} \cup \{a\} \cdot \bigcup_{k \geq 0} \{bc\}^k$$



Syntax

language union

Kleene closure

$$abba + a \cdot (bc)^*$$

every word is a regular expression

language concatenation

Definition: The set *RegEx* of **regular expressions** E is given by:

$$E \ ::= \ \emptyset \mid \varepsilon \mid a \mid E_1 + E_2 \mid E_1 \cdot E_2 \mid E_1^*$$

symbol in Σ

$\cdot$ is typically omitted

Order of precedence: $* \ > \ \cdot \ > \ +$

Semantics of Regular Expressions

Definition: The **language-based semantics** of regular expressions is a function

$$L: \quad RegEx \quad \rightarrow \quad Powerset(\Sigma^*)$$

that is defined recursively on their syntax.

set of all languages over Σ

$\boldsymbol{E}$	$\boldsymbol{L(E)}$
$\emptyset$	$\emptyset$
ε	$\{\varepsilon\}$
a	$\{a\}$

$\boldsymbol{E}$	$\boldsymbol{L(E)}$
$E_1 + E_2$	$L(E_1) \cup L(E_2)$
$E_1 \cdot E_2$	$L(E_1)\ L(E_2)$
E_1^*	$L(E_1)^*$

Examples from Compilers

Keywords in programming languages

```
K  =  if  +  else  +  while  +  do

L(K) = { if, else, while, do }
```

Identifiers: `L(id)` describes all alpha-numeric sequences that start with a letter

```
id    =  alpha (alpha + digit)*

alpha  =  a + b + c + … + z + A + B + C + … + Z

digit  =  0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9
```

What languages are described by the following expressions?

1. $(a + b)^* a (a+b) (a+b)$

2. $(a + b)^* (a + bb)$

3. $(a+b)^* aa (a+b)^*$

Applications of Regular Expressions

Regular Expressions in Practice

- Unix tools: grep, awk, perl, …
- Text editors: *VSCode*, vim, emacs, …
- Compilers: define keywords, names, …

- All of these tools use similar syntax with lots of additional syntactic sugar

Our RegEx.	Extended RegEx.
a + b	a | b
a + ε	a?
a a*	a+
a+b+c	[abc] [a-c]

more on regular-expressions.info

Exercise: RegEx Golf

Find a regular expression that matches all words on the left-hand side and none of the words on the right-hand side.

Note: “matches” here means that a word contains a word described by the regular expression as a sub-string.

Match all of these…	and none of these…
✔ afoot	✘ Atlas
✔ catfoot	✘ Aymoro
✔ dogfoot	✘ Iberic
✔ fanfoot	✘ Mahran
✔ foody	✘ Ormazd
✔ foolery	✘ Silipan
✔ foolish	✘ altared
✔ fooster	✘ chandoo
✔ footage	✘ crenel
✔ foothot	✘ crooked
✔ footle	✘ fardo
✔ footpad	✘ folksy
✔ footway	✘ forest
✔ hotfoot	✘ hebamic
✔ jawfoot	✘ idgah
✔ mafoo	✘ manlike
✔ nonfood	✘ marly
✔ padfoot	✘ palazzi
✔ prefool	✘ sixfold
✔ sfoot	✘ tarrock
✔ unfool	✘ unfold

Open Questions:

Can every language accepted by an automaton also be described by a regular expression?

Can every language described by a regular expression also be accepted by a finite automaton?

Yes!

that's why we say DFAs accept *regular languages*

From Regular Expressions to Finite Automata

Main idea: construct an ε-NFA $A(E)$ for every regular expression E

“Thompson's construction”

Exercise

Try to construct an NFA for the regular expression $(0+1)^* \, 101 \, (0+1)^*$

Automata-Based Semantics of Regular Expressions

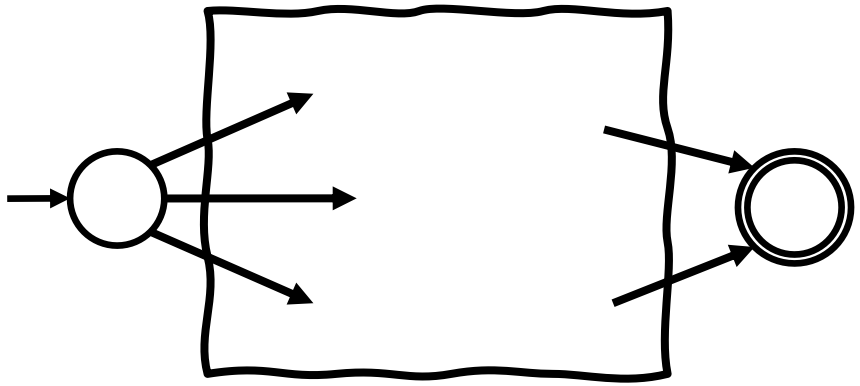
Definition: The **automata-based semantics** of regular expressions is a function

$$A: \quad RegEx \quad \rightarrow \varepsilon\text{-NFAs}$$

that is defined recursively on their syntax.

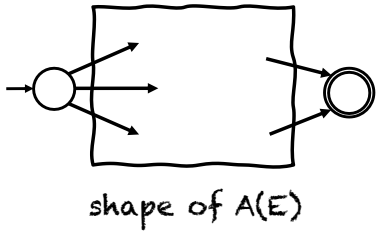
We construct the ε-NFA $A(E)$ such that it maintains three properties:

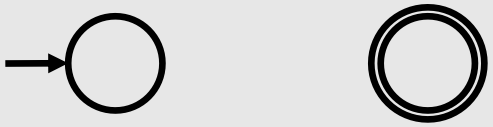
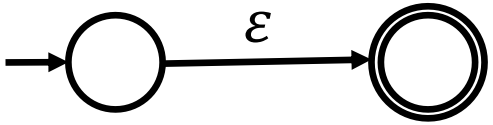
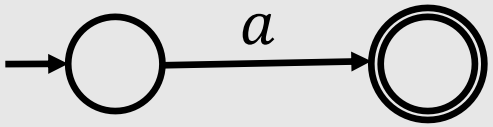
1. $A(E)$ has exactly one final state
2. $A(E)$ has no transitions that enter the initial state
3. $A(E)$ has no transitions that leave the final state

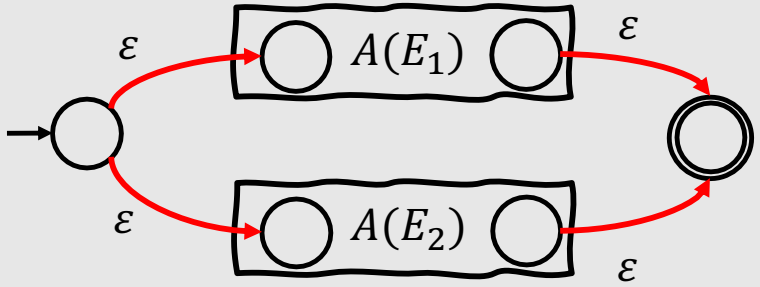
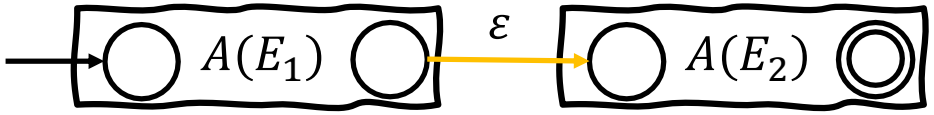
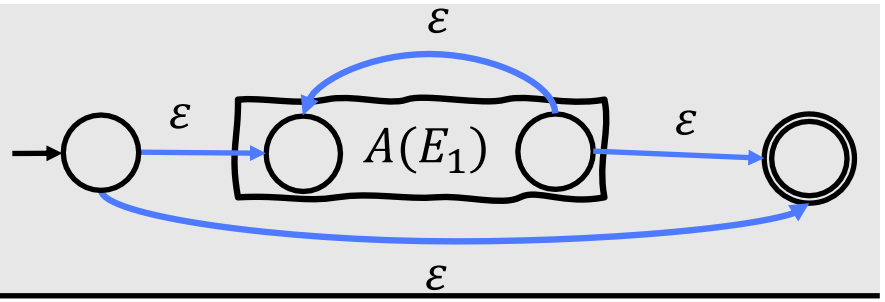


shape of A(E)

Automata-Based Semantics of Regular Expressions



E	$A(E)$
$\emptyset$	
ε	
a	

E	$A(E)$
$E_1 + E_2$	
$E_1 \cdot E_2$	
E_1^*	

Example: an automaton for 1 0*

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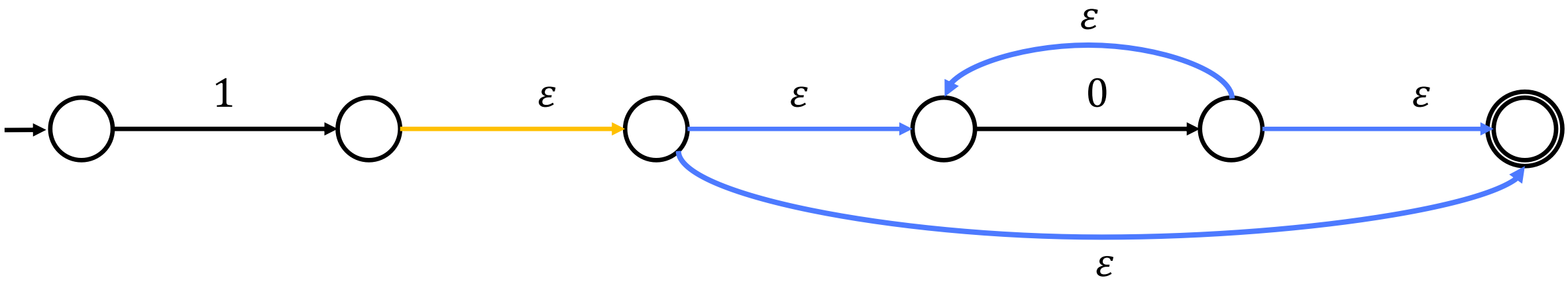


Regular expressions accept regular languages

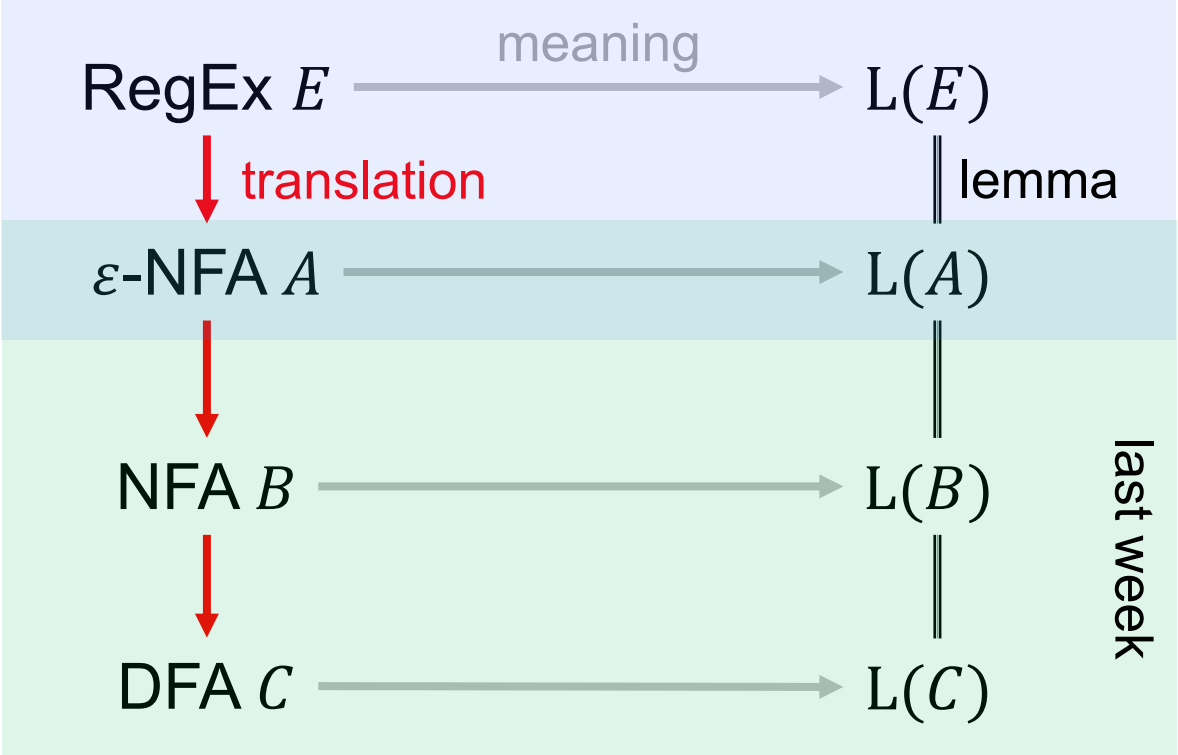
Lemma (semantic equivalence).

For every regular expression E, we have $L(E) \;=\; L(\,A(E)\,)$.

Proof. By structural induction on the syntax of E. (on the board)

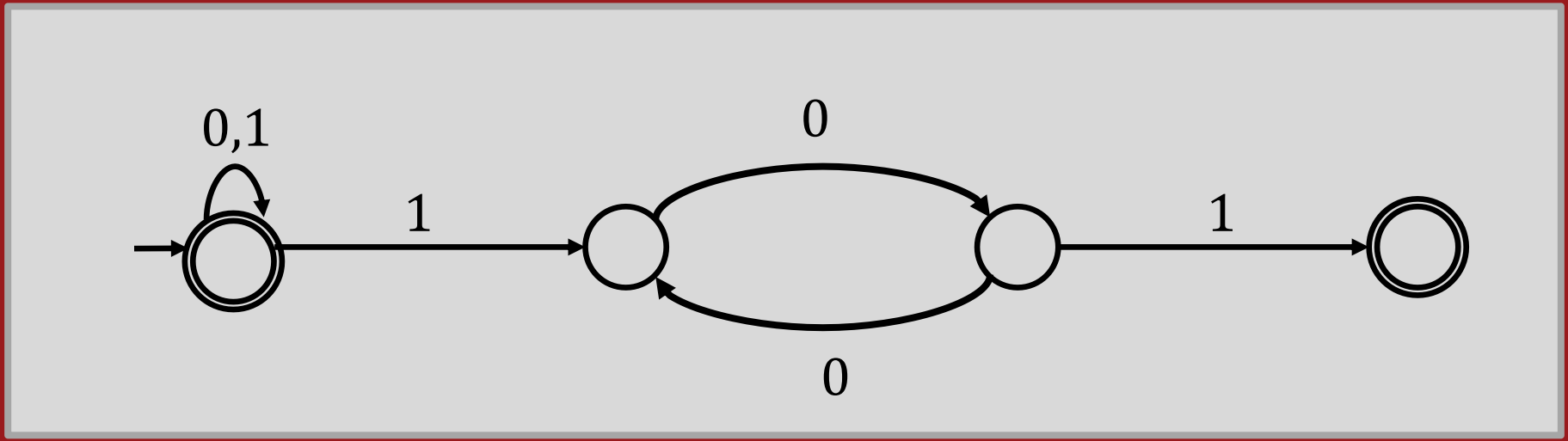
Theorem.

For every regular expression E, there exists a DFA C such that $L(E) = L(C)$.



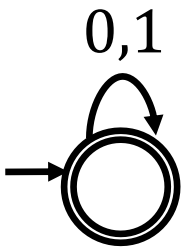
From Finite Automata to Regular Expressions

Exercise: Try to construct a RegEx for the following NFA



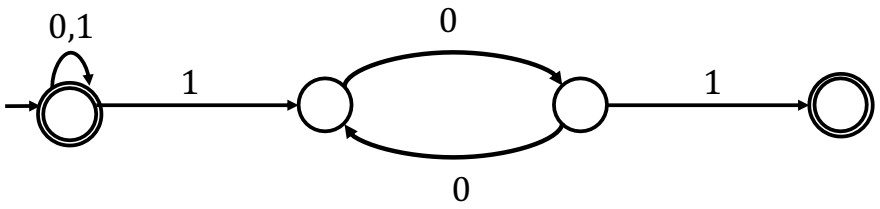
Insights

1. For small NFAs with one final state, we can construct a RegEx according to a pattern.



(0+1)*

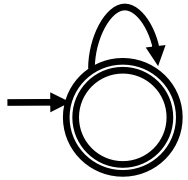
2. We can treat multiple final states as a sum of NFAs with one final state.



(0+1)* + (0+1)* 1 0 (00)* 1

3. If the alphabet consists of regular expressions, then the RegEx obtained from a pattern is a RegEx over the symbols in the "RegEx alphabet".

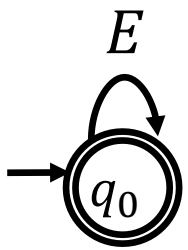
$a\ b^*\ a$



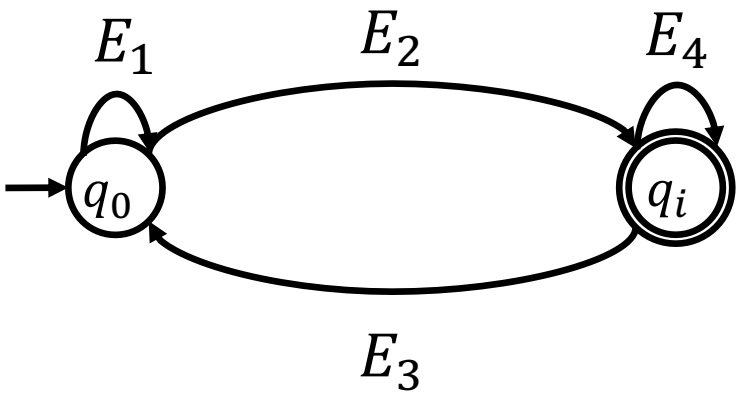
$(a\ b^*\ a)^*$

➔ Idea: reduce NFAs to known patterns by eliminating states and labeling edges with RegEx's.

NFA to RegEx: Fixed Patterns

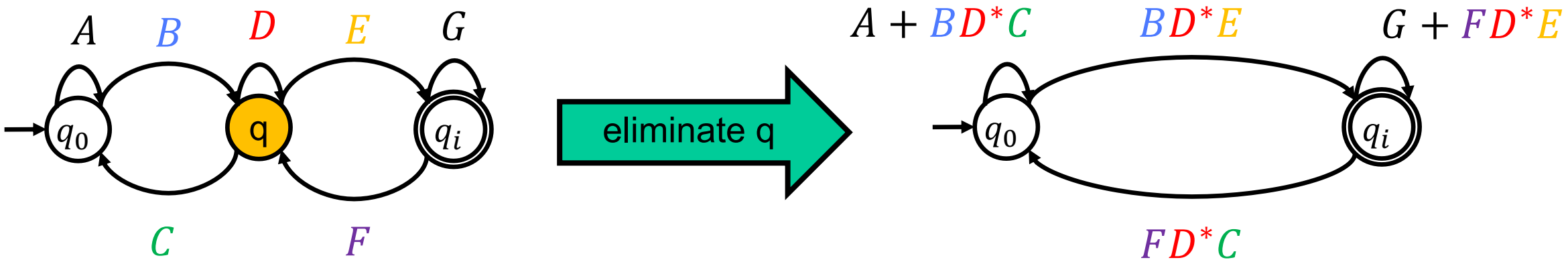


E^*



$E_1^* \; E_2 \; (E_4 + E_3 \, E_1^* \, E_2)^*$

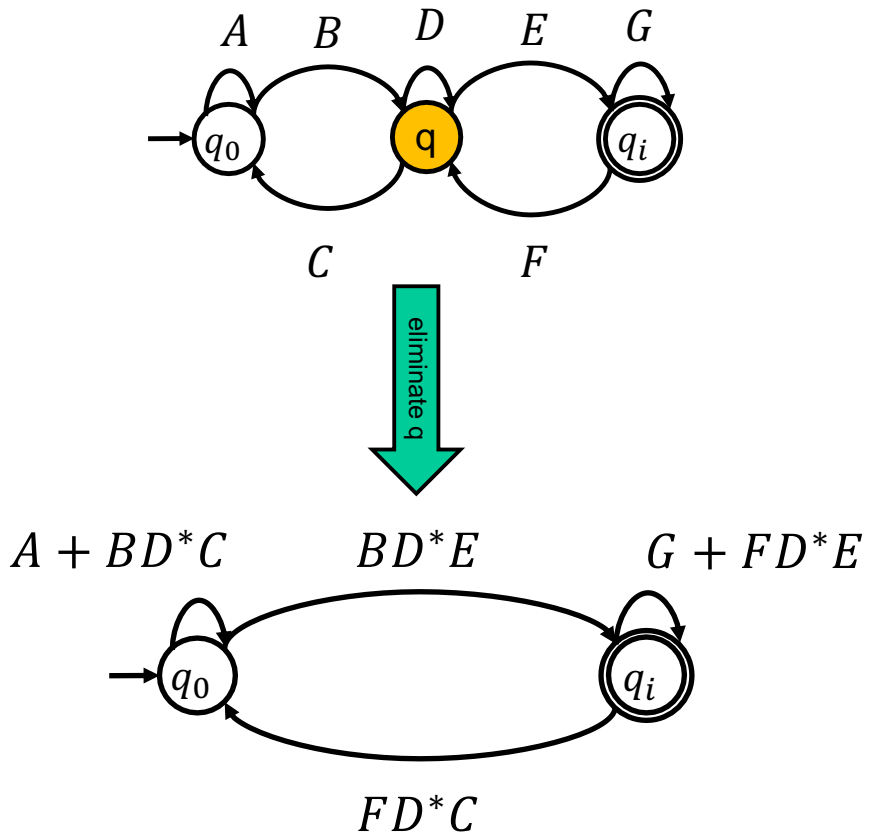
NFA to RegEx: State Elimination



NFA to RegEx: Algorithm

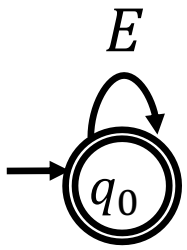
Input: NFA A with initial state q_0 and final states F.

Output: regular expression E such that $L(E) = L(A)$.

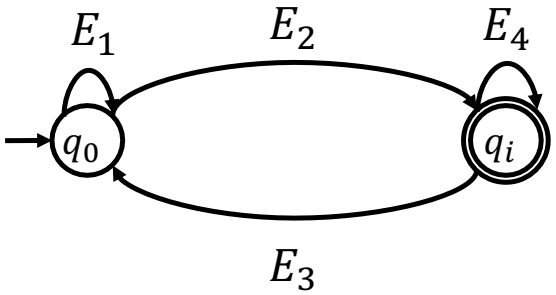


1. For every final state $q_i \in F$:
   i. Eliminate all states of A except q_0 and q_i

   ii. Derive a regular expression R_i by pattern matching



$R_i = E^*$

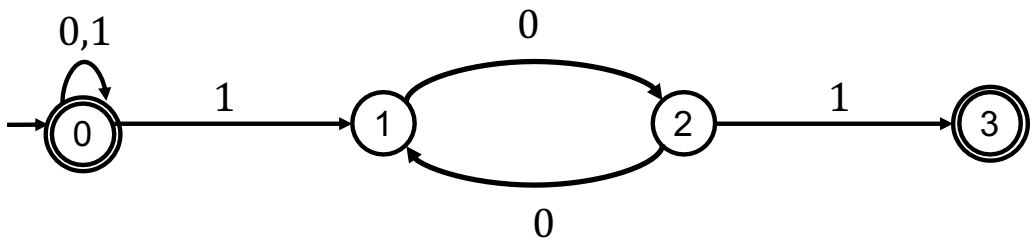


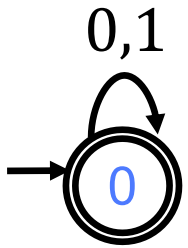
$R_i = E_1^* \; E_2 \; (E_4 + E_3 \, E_1^* \, E_2)^*$

2. Return the sum of all derived regular expressions

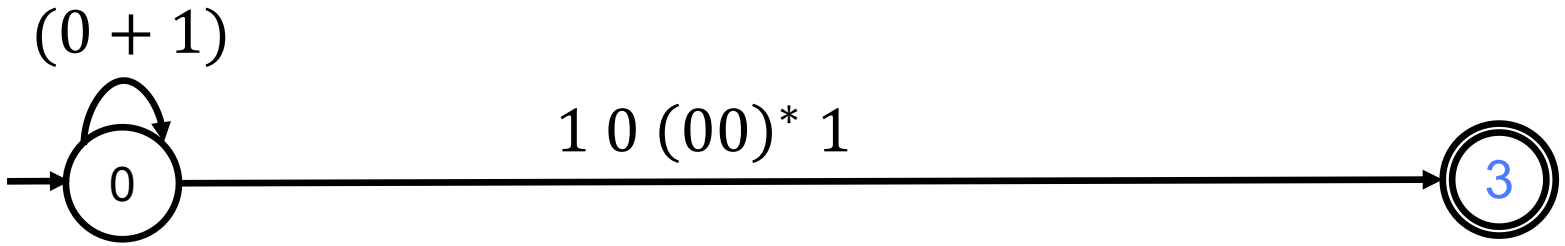
$$E = \sum_{i \; s.t. \; q_i \in F} R_i$$

Example





$$R_0 = (0+1)^*$$



$$R_3 = (0+1)^*\ 1\ 0\ (00)^*1$$

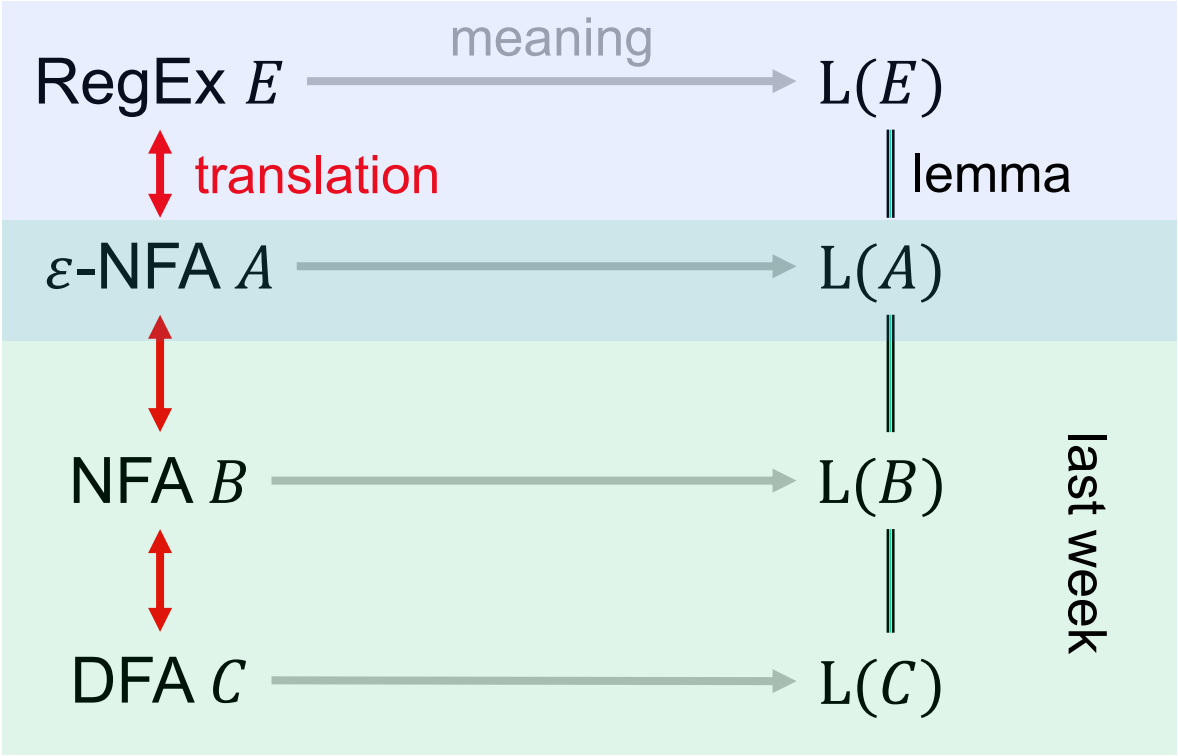
$$E \ = \ R_1 + R_3 \ = \ (0+1)^* + \ (0+1)^*\ 1\ 0\ (00)^*1$$

Summary

- Regular expressions and DFAs accept exactly the class of regular languages

RegEx ➔ DFA via Thompson's construction

DFA ➔ RegEx via state elimination



Quiz

1. Given two regular expressions E_1 and E_2, is there always a regular expression E such that $L(E) = L(E_1) \cap L(E_2)$?

2. Can we check whether all words accepted by a DFA A are also in the language of a regular expression E?

3. Is there a regular expression that is significantly smaller than any DFA accepting the same language?

Properties of Regular Expressions

Equivalence of Regular Expressions

Definition: Regular expressions are **equivalent** if they have the same language

$$E_1 = E_2 \quad \text{if and only if} \quad L(E_1) = L(E_2)$$

$$\left(\left(0\,(1+\varepsilon)\right)^* + (0+01)^*\right)^* \quad = \quad (01+0)^* \ ?$$

- Approach 1: transform both into DFAs and check equivalence (later)
- Approach 2: simplify regular expressions using calculation rules

„language union and concatenation behave similarly to addition and multiplication“

Regular Expression Laws

A law is an equality $E = R$ between regular expressions E, R with unknowns (e.g. X, Y, Z) such that we obtain an equivalence whenever we replace the unknowns by regular expressions.

Examples:

- Distributivity law: $X\,(Y + Z) = \; X\,Y + X\,Z$
- Commutativity law for union: $X + Y \; = Y + X$
- Identity law for concatenation: $X \cdot \varepsilon \; = \varepsilon \cdot X \; = \; X$

Quiz: Determine which rules are laws for regular expressions

1. Associativity for addition: $(X + Y) + Z = X + (Y + Z)$

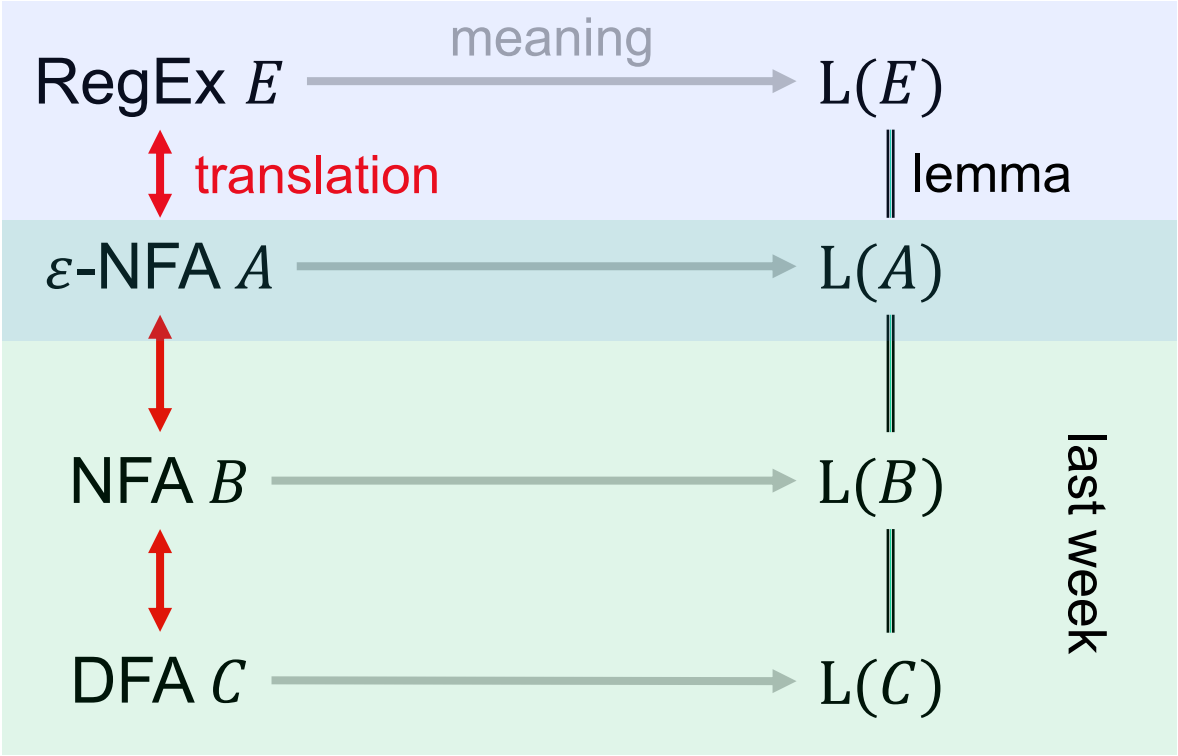
2. Commutativity for concatenation: $X\,Y = Y\,X$

3. $(X^*)^* = X^*$

4. $\emptyset^* = \varepsilon$

Wrap-up: Regular Expressions

- Regular expressions and DFAs accept exactly the class of regular languages
  - RegEx ➔ DFA via Thompson's construction
  - DFA ➔ RegEx via state elimination
- Regular expressions are *declarative*
  - Often concise descriptions of languages
  - Hard to solve problems like membership
  - Enjoy many algebraic properties of + and ·
- DFAs are *operational*
  - Easy to solve problems like membership
  - DFAs can be exponentially larger than regular expressions for the same language



What next?

- So far: formal models for regular languages
  - Well-behaved setting
  - But: limited expressiveness

- Friday: a formal model for running *programs*
  - ➔ Foundation for mathematical reasoning about programs
  - ➔ Code verification
  - ➔ Optimization
  - ➔ Compiler checks
  - ➔ IDE support

Exercise Class

- Remember: *strict* deadline for group formation: 16 February
- Today's exercises: DTU Learn > Content > RL3 > exercises.pdf
  - You can always ask us for feedback and/or help
  - Solutions will be published on Thursday
- **Strongly recommended:** submit your solution to the marked "Homework" task
  - **When:** before Wednesday 12:00
  - **Where:** DTU Learn > Assignments
  - Tasks are similar to exam tasks
  - This is your chance to get detailed written feedback
  - Poorly written or informal solutions are a major reason for losing points in the exam

Thanks for today!